

IMC Prosperity 4

Algorithmic Trading Strategy

Edgar Cornillet — Team Lead

Télécom SudParis — Institut Polytechnique de Paris

Round 1 Result: 100,994 XIRECs — Rank 325 (Algorithmic Leaderboard)

Competition Context

IMC Prosperity 4 is an international algorithmic trading competition organized by IMC Trading. Participants design Python algorithms to trade products on simulated real-time markets, competing against bots with various behaviors. The goal is to maximize Profit & Loss (PnL) across multiple trading days.

In Round 1, two products were available: **Intarian Pepper Root** (position limit: 80) and **Ash Coated Osmium** (limit: 80). We had access to 3 days of historical data for similar products.

Baseline Architecture: the V3 Market Maker

Market Making Principle

A market maker provides liquidity by simultaneously posting buy orders (*bids*) and sell orders (*asks*). Profit comes from the spread between buy and sell prices. It is a non-directional strategy: we profit from order flow, not from market direction.

Technical Components

The V3 algorithm was built on five pillars:

(a) **Fair Value via EMA.** We estimate fair value through an Exponential Moving Average of the volume-weighted mid price (WMID). An alpha of 0.15 balances responsiveness and stability.

(b) **Inventory Management via Exponential Skew.** Fair value is adjusted based on current position using an exponential formula:

$$\text{skew} = \frac{\text{position}}{\text{limit}} \times \left(1 + 2 \times \left(\frac{\text{position}}{\text{limit}} \right)^2 \right)$$

The penalty is mild for small positions but spikes near the limits, preventing extreme exposure.

(c) **Multi-Level Quoting.** Orders are posted at two price levels. Level 1 (60% of budget) uses the minimum of the *penny* ($\text{best_bid} + 1$) and the theoretical price. Level 2 (40%) posts 1 tick deeper.

(d) **Volatility-Adaptive Spread.** The base spread widens with volatility and tightens when the market is calm, measured via an EMA of absolute price changes.

(e) **Aggressive Taking Layer.** When the order book shows a clearly mispriced level relative to our fair value, we take liquidity immediately.

V3 Performance: ~4,883 on the test sample.

Log Analysis: the Methodological Breakthrough

The real progress came from quantitative analysis of **simulation logs**. Instead of modifying the algorithm based on intuition, we extracted insights from every trade and every tick.

Discovery 1: Real Spreads Are 5× Wider

Historical data showed spreads of 2–3 ticks. Actual data: Pepper Root at 13, Osmium at 16.

Discovery 2: Spike Bots Are Informed

We analyzed all 71 *spikes* (moments when bid/ask jumped by >5 ticks). Measuring PnL 5 ticks after each spike trade:

Average PnL per spike trade: −1.1 — Only 33% profitable

Bots placing aggressive orders *knew* the price was about to move in their direction. Trading against them was a losing proposition. This explained why aggressive versions (V4, V5, V6) lost money: they caught more spikes, but each spike had negative expected value. **Key concept: adverse selection.**

Discovery 3: Pepper Root Has a Constant Drift

Pepper Root showed a linear, constant upward drift of +101 ticks over 100,000 ticks (+52 first half, +49 second half). Osmium was perfectly stable around 10,000.

Discovery 4: Only 12% Position Utilization

Maximum position reached: ~10 out of a limit of 80. The bottleneck was fill frequency (142 trades in 100,000 ticks), not capacity.

The Innovation: Trend Capture with Confidence Gate (V9d)

Concept

On a trending product, we capture directional movement **on top of** market making profit by biasing fair value in the trend direction:

$$\text{adj_fair} = \text{fair} + \text{imbalance} \times w_{\text{imb}} - \text{skew} \times w_{\text{skew}} + \underbrace{\text{trend_bias}}_{\text{new}}$$

Trend Detection

Two EMAs at different speeds:

- **Fast EMA** ($\alpha = 0.15$): tracks price closely
- **Slow EMA** ($\alpha = 0.02$, ~50 tick memory): smooths out noise

The trend signal is the difference: $\text{trend} = \text{EMA}_{\text{fast}} - \text{EMA}_{\text{slow}}$.

The Confidence Gate (Key Innovation)

We avoid reacting to noise by computing **trend confidence**:

$$\text{trend_confidence} = \min\left(1, \frac{|\text{trend}|/\sigma}{2}\right)$$

where σ is estimated volatility. Both bias and skew adapt dynamically:

Regime	Confidence	Bias	Skew	Behavior
Strong trend, calm	1.0	$\times 7.0$	1.5	Aggressive, accumulates position
No trend	0.0	$\times 0.0$	3.0	Falls back to V3 baseline
Weak trend, choppy	0.25	$\times 1.2$	2.6	Cautious

Score Progression

Version	Score	Change	Key Change
V3 (baseline)	4,883	—	Generic market maker
V9 ($\times 3$)	5,775	+18%	First trend bias
V9b ($\times 5$, skew=2)	7,664	+57%	Stronger bias
V9c ($\times 7$, skew=1.5)	9,202	+88%	Maximum bias
V9d ($\times 7$ + gate)	9,202	+88%	Same perf + choppy market protection

V9d is identical to V9c on a trending market but automatically protects on a trendless one. This is the version we submitted. **Final competition result: 100,994 XIRECs.**

Overfitting Risk Management

The main risk was calibrating parameters on a single sample. Our protections:

- **The confidence gate:** without a detected trend, the bias drops to 0, reverting to V3 ($\sim 4,800$ guaranteed).
- **Per-product separation:** Osmium has zero trend bias (identical to V3). Only Pepper Root has trend capture.
- **Qualitative validation:** in-game tips confirmed “slow growth creates structure” for Pepper Root.

Manual Challenge: Auction Optimization

We coded an **auction simulator** testing every possible (price, volume) pair. For Ember Mushroom, the key insight: by placing a large buy order (35,000 units), we **shift** the clearing price from 15 to 18 while remaining below the buyback price (20). Profit: 66,500.

Key Takeaways

1. **Data over intuition.** Every decision was validated by data. “Clever” versions all lost money. Log analysis revealed that spike bots were informed — a discovery that shaped the entire strategy.

2. **Adverse selection is the market maker's real enemy.** Being more aggressive isn't better. The conservative V3 beat every aggressive variant because it avoided trading against informed bots.
3. **Conditional risk management beats static risk management.** The confidence gate dynamically adapts risk to the market regime — more sophisticated than a fixed stop-loss.
4. **Microstructure determines strategy.** The same products on different markets (spread of 3 vs 16) require fundamentally different approaches.
5. **Team leadership under pressure.** As team lead of 4, I coordinated analysis, strategic decisions, and version selection under tight 48-hour deadlines per round.